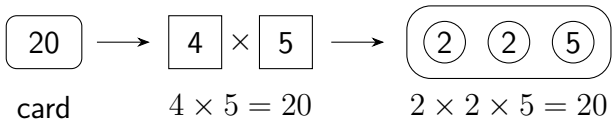
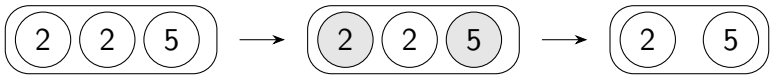
A number card can split into two whole numbers greater than 1 that multiply to its value. These numbers are its factors. Keep splitting until every card is prime: a number greater than 1 that cannot split. The finished cards make a prime bag. Multiply its tokens to find its value.



Token order does not matter. Tokens with the same number are the same.

Choose some tokens from a bag, using each token at most once. The chosen tokens form a bag. You may choose all of them or none. An empty bag has value 1. A divisor of a whole number divides it with no remainder.



value 20 choose shaded tokens value 10

$2 \times 5 = 10$ $20 \div 10 = 2$

Problem 4: Use prime tokens with any of the labels 2, 3 and 5. Build a bag that makes exactly 4 different values by choosing tokens, another that makes exactly 6, and another that makes exactly 8. How small can you make the full bag's value in each case?

4 different values

6 different values

8 different values